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Self, Culture, and Society III

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Due Mar 24th, 2023

The Anthropocene: A Budding Scientific Revolution

Our earth's history can be divided up into epochs, a unit that helps make up the geologic time scale. Officially, we are living in the Holocene epoch, a period that encompasses all passed time since the last ice age and all of recorded human history. In "Postcolonial Studies and the Challenge of Climate Change," Dipesh Chakrabarty writes about humans in the "Anthropocene," an epoch defined by the period in which humans act as a geological force on the planet. The relatively recent coining of the Anthropocene –introduced in 2000 by chemist Paul Crutzen– implies a departure from the Holocene epoch, but the International Union of Geological Sciences, who have the official authority to name and define epochs, has not recognized the Anthropocene as any different from the Holocene. While the term was only proposed recently, the Anthropocene is said to describe a time period dating back to about the 1950s. If in the 1950s, supposedly a part of the Anthropocene, scientists were studying the climate under the presumption that they were in the Holocene, and we were able to develop climate science as it exists today, is there anything really different about the reality between the two aforementioned epochs? Or, do the two "time periods" assume fundamentally different things about the world that allows us to view the same point in time in completely different ways? Viewing the Anthropocene as a scientific paradigm, as defined by Thomas Kuhn in *The Structure of Scientific Revolutions*, helps us understand how shifts in how we think about our world can change what we observe as scientifically and socially relevant or important. While the state of the climate in

the proposed Anthropocene is drastically different from previous parts of the Holocene and could warrant the definition of a new epoch, the Anthropocene's most prominent trait is the recognition of human impact on the earth; arguably, viewing human activity as having dominant control over earth's climate and environment entails a fundamental shift in the practice of science and can be best described as the introduction of a new scientific paradigm.

To explain why the Anthropocene is more of a scientific paradigm than a new period, we must first look at what constitutes a scientific paradigm. In *The Structure of Scientific Revolutions,* Thomas Kuhn discusses how the history of science is driven by paradigm shifts. Kuhn describes paradigms as being able to "define the legitimate problems and methods of a research field for succeeding generations of practitioners" (Kuhn, 10). Furthermore, paradigms lay a foundation for scientific standards and consensus in a scientific community, making the practice of "normal science" possible. Normal science, "the activity in which most scientists inevitably spend almost all their time, is predicated on the assumption that the scientific community knows what the world is like" (Kuhn, 5, 11). The way our international scientific community understands the world today is as existing in the Holocene. In general, the Holocene represents a warmer period of the earth's history in between ice ages. The Holocene has seen natural disasters, the rise of mankind, and the transition towards urban and technological living. We have not entered another ice age and the characteristics of the world today seem to still fit within those of the Holocene. Simply put, the world we live in has been scientifically understood as part of the Holocene for thousands of years – normal science lives within the Holocene. But, climate change and the relatively rapid global warming of the last century have presented climate scientists with new challenges. Climate science itself only began garnering serious attention in the mid-20th century. While the Holocene is characterized as a "warmer" period, just *how* warm

the epoch we are living in is the main subject of climate science today. The unprecedented changes in the climate in the last century compared to the last hundreds of thousands of years have proposed new areas of study for scientists and can be described as "anomalies" in the normal science of the Holocene. Today, upon recognizing these anomalies, we have begun to call global warming the "climate crisis", a crisis that scientists attribute to the human burning of fossil fuels. Kuhn, when discussing historical paradigm shifts, describes how the progression of anomaly to crisis is a common characteristic of most paradigm shifts: "in all these cases except that of Newton the awareness of anomaly had lasted so long and penetrated so deep that one can appropriately describe the fields affected by it as in a state of growing crisis" (Kuhn, 67). The climate crisis seems to closely resemble Kuhn's idea of a paradigmatic shift. Climate scientists started to recognize that natural disasters, climate events, and rising sea levels were not just due to naturally occurring changes in greenhouse gas concentrations or natural changes in our atmosphere or the sun's intensity – there is an increasing sentiment within the scientific community and general public that to blame the increase in flooding, air pollution, and other extreme weather events on natural changes in the climate would be ignorant of human's presence and impact on the earth. Attributing changes to the climate to something other than a natural process has required new research (Chakrabarty dates the scientific idea that humans are a geophysical force back only to 1957) and the further development and reinvention of climate science (Chakrabarty, 10) We still don't know the full extent to which humans impact the climate, but the recognition that we may need to know is exactly what the coining of the Anthropocene proposes we need to explore.

For Kuhn, paradigms that address anomalies and lead to scientific crisis demand scientific revolution: "it demands large-scale paradigm destruction and major shifts in the

problems and techniques of normal science, the emergence of new theories is generally preceded by a period of pronounced professional insecurity…that insecurity is generated by the persistent failure of the puzzles of normal science" (Kuhn, 67-68). The adoption of the Anthropocene, with its focus being the climate crisis, closely follows Kuhn's description of paradigmatic revolution. It demands a change in the way we study our environment and climate, our current paradigm, and shifts in the techniques of today's normal science are already happening in the fields of geology, engineering, and meteorology. The insecurity Kuhn describes has been displayed by the IUGS's hesitation to adopt the Anthropocene as a new epoch due to a lack of rock strata data as well as the abundance of climate change deniers across the globe. But, even if the rock strata don't contain the data to prove that humans have changed the earth, the very fact that techniques are being developed to conduct research to prove this shows that, even if the Anthropocene is not deemed an epoch, it marks a new way of scientific thinking, a paradigm shift. Climate change deniers are often met with relentless scrutiny on social media with hypothetical arguments like, "if climate change wasn't caused by humans then how do you explain….". Scientific arguments that refrain from factoring in human existence in the study of the natural world are seen as ignorant and sometimes deemed conspiracy because their "normal science" fails to answer new puzzles addressed by the paradigm shift. One puzzle popping up in the normal science of today is the rapid loss of biodiversity and the predicted loss of three-fourths of the globe's mammal species. These numbers have caused scientists to believe we are on the brink of the sixth mass extinction in Earth's history. But, unlike previous mass extinctions, which were caused by natural phenomena, this mass extinction will have been driven by human activity. While scientists predict that the sixth mass extinction is probable, it is not inevitable because of its nature as a human-caused event. This, a preventable mass extinction, is unprecedented, and some biologists

are already studying how to protect threatened mammalian species. The way scientists think about and develop technology to address conservation and wildlife protection is changing and aligns with the ideas proposed by the Anthropocene. More and more, it seems that the Anthropocenic paradigm is in the process of revolutionizing science.

Some may argue that we already live in the Anthropocene, and the introduction of the term merely gives a name to the world we live in and the science we practice, but, continuing to view the Anthropocene as a paradigm, we can see that the scientific revolution towards the Anthropocene has started but is not complete. The paradigm shift to the Anthropocene will be complete when the substantial impact humans have on the planet is seen as "a new basis for the practice of science" (Kuhn, 6). In other words, the Anthropocene must become the new normal science for us to recognize it as our current paradigm. Right now, science that explores their fields with Anthropocenic ideas at its core is what Kuhn calls "extraordinary science". Chakrabarty discusses how scientists today are conducting this extraordinary science: "They are also giving us an account of climate change that is neither purely "natural" nor purely "human" history. And this is because they assign an agency to humans at the very heart of this story.…humans are now part of the natural history of the planet" (Chakrabarty, 10). Scientists are beginning to reimagine the way the climate is studied which, according to Chakrabarty, has consequences for the human sciences which have traditionally been constructed by a separation between natural and human history. This extraordinary reimagination is beginning to make its way into textbooks and education systems, the home of normal science, which bodes well for the fate of the Anthropocene as an adopted paradigm. If the Anthropocenic paradigmatic revolution is successful, human agency and the natural sciences of the planet will ideally fuse as a single

subject in people's minds and become scientific knowledge – "intrinsically the common property of a group", according to Kuhn – rather than a theory or anomaly (Kuhn, 210).

The Anthropocene not only calls into question established modes of studying the climate but also the broader involvement of the human in the environment. Chakrabarty references a quote from climate researcher Mike Hulme: "This global solution-structure also begs a fundamental question…What is the ultimate performance metric for the human species, what is it that we are seeking to optimise?" (Chakrabarty, 13). Hulme goes on to list questions about humans' motives for discussing policy around climate change. The Anthropocene's central claim is that humans are a geophysical force on the planet and the field of climate science has become extremely politicized. While this may seem like resistance to the Anthropocene as a paradigm, it can actually be viewed as support, no matter where the political characterization is positive or negative. Either way, humans deciding whether or not they need to make policy regarding climate change is an indicator that humanity has started to embrace their connection to the environment and the fate of the world. People that believe climate change is not a human issue implicitly make the argument that humans are a part of nature and damage done to the climate during our time on earth is natural; people who believe climate change *is* a human issue enforce the anthropocenic human-nature connection similar to the researchers that believe we can halt the acceleration towards the sixth mass extinction. Furthermore, a sect of climate science is finding solutions and creating technologies to slow climate change rather than simply researching the state of the climate. Human intervention is baked into the study and remediation of climate change because, ultimately, climate change is a threat to humanity's survival. Kuhn points out that, historically, this practical importance, human survival, of the Anthropocene bodes well for its success as a scientific revolution: "as in the Copernican revolution, an anomaly

without apparent fundamental import may evoke crisis if the applications that it inhibits have a particular practical importance" (Kuhn, 82). The unique importance this human-nature connection has in the context of climate change also becomes an attractive concept for scientists to delve into which Kuhn says is an essential characteristic of emerging paradigms: "Their achievement was sufficiently unprecedented to attract an enduring group of adherents away from competing modes of scientific activity" (Kuhn, 10). Ultimately, the nature of the Anthropocene as establishing a connection between natural science and human history and behavior makes it a strong candidate for scientific revolution.

While the Anthropocene has many elements that make it a strong candidate for successful paradigmatic revolution, adjustment to the new paradigm does not happen immediately. Speaking on scientists' adjustment to new paradigms, Kuhn cites Max Planck: "a new scientific truth does not triumph by convincing its opponents and making them see the light, but rather because its opponents eventually die, and a new generation grows up that is familiar with it" (Kuhn, 151). If Planck is right about the natural adoption of the paradigm with time, then the shift to the Anthropocene seems likely; younger generations today are growing up learning about the climate crisis and have helped proliferate the global climate action movement because of climate change's potentially fatal impact on their futures. But, the climate emergency and need for climate solutions is urgent and the gradual acceptance of the climate crisis is popularly criticized in climate activism; this has caused uproar from climate crisis deniers, making the minority opinion just as loud as the majority. It will be interesting to see if the adoption of the Anthropocene as an epoch accelerates its acceptance as a paradigm or whether the shift will happen regardless of its status as a geochronological unit. Ultimately, the Anthorpocene's credentials as a paradigm lead me to predict the latter will happen.

Works Cited

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